Stage 1: Development

CWRU and Paderborn only; target data sealed

CWRU

Paderborn

Pipeline built and fixed

envelope-order spectra, 7 network arms
and a train-free baseline (C0)
gate G1 passed on both datasets

Dry run and power analysis

every arm on CWRU to Paderborn and back
minimum detectable difference simulated

Target datasets downloaded, kept sealed

UORED-VAFCLS KAIST HUST

label-blind export to one file, SHA-256 checksum
read only: manifest, array lengths, UORED speed column

Development log (DEV_LOG.md)

every recipe change dated before registration
the speed-column read declared in advance
G1 history and dry-run results kept
for reporting in full

Stage 2: Preregistration

Protocol, code and checksums frozen

Frozen before opening

hypotheses H1 to H3
metric: per-bearing Brier score
test: exact sign-flip, Holm
quality gates G2 to G4
code: commit 331fdff
tag: prereg-freeze
data: SHA-256 checksums

OSF registration

osf.io/qsh2a

2 November 2026

Update after opening

deviation D1 declared
two fold-handling fixes
gate results posted
exploratory additions
listed separately

Stage 3: Target data opened

Gates checked before any confirmatory run

UORED-VAFCLS

KAIST

HUST

Gates

G2: diagnosable by envelope?
G3: isolation, checksums, speed check
G4: shuffled labels

UORED shaft-speed check (G3)

stored speed matched the vibration
spectrum in only 21 of 40 recordings
stop and correct: label-blind
spectral rule registered for KAIST

Deviation D1 logged

corrected speed used in main analysis
stored speed kept as sensitivity run
envelope check supports correction
(16 of 18 differing recordings)
conclusion unchanged under both

Stage 4: Analysis and reporting

Every registered run reported

Confirmatory (registered)

Paderborn to UORED-VAFCLS
10 physical test bearings

H1	M3 vs M2	n.s.
H2	M4 vs M5	n.s.
H3	M4 vs C0	n.s.

Descriptive

12 directed transfer pairs
HUST leave-one-type-out (5 folds)
17 settings, every arm reported

Controls and sensitivity

shifted and random fault-order centres
shuffled-label control
stored UORED speed
model-free rig and fault analysis

Exploratory (not registered)

WDCNN raw-signal network
per-arm hyperparameter search
labelled exploratory;
no hypothesis test

Throughout: every physical bearing stays on one side of every split, and the code asserts this in every run.